

Sergio Eduardo Vargas López

Senior Backend Engineer - Remote (Worldwide)

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Summary

Backend-focused Software Engineer with 7+ years of experience building scalable APIs, event-driven services, and serverless systems. Strong background in Python (FastAPI, Django, Flask), AWS (Lambda, RDS, CloudWatch, Secrets Manager), and PostgreSQL performance optimization. Experienced in designing reliable production systems, improving test coverage, and collaborating across teams to deliver secure and maintainable software. Recent work includes AI/LLM orchestration, evaluations, and production integrations using LangGraph, LangSmith, and modern LLM tooling.

Selected Achievements

- Increased automated test coverage by **35%** (integration/unit) for mission-critical services.
- Improved complex PostgreSQL queries and indexing for approximately **25% faster** response times.
- Supported payment processing systems handling **50,000+** monthly transactions with high reliability.
- Automated Linux server operations, reducing manual workload by approximately **40%**.

Experience

Unosquare / Harvard Business School - Senior Backend Engineer

Nov 2022 - Present

- Designed and delivered Python/FastAPI services for production use, building scalable and maintainable backend systems.
- Built AWS serverless components using Lambda, RDS, CloudWatch, and Secrets Manager to support secure deployments and operational visibility.
- Led requirements discussions and translated user needs into technical specifications and acceptance criteria.
- Drove testing strategy across integration and unit tests, increasing coverage by 35% and reducing regressions.
- Implemented LangGraph and LangSmith workflows for orchestration, tracing, and evaluations in AI/LLM-powered features.
- Integrated OpenAI and Claude-based flows with streaming, guardrails, and retrieval/action routing patterns.

EON Igniting Business - Backend Developer

Jul 2021 - Nov 2022

- Developed microservices in Python/Django using hexagonal architecture.
- Tuned PostgreSQL performance through query plan analysis, indexes, and connection handling, achieving approximately 25% latency reduction.
- Coordinated with QA, product, and frontend teams in Scrum to deliver reliable backend features on time.
- Integrated PostgreSQL with Python using SQLAlchemy and psycopg2.

Paymentez - Backend Developer

Oct 2019 - Jul 2021

- Built Django/DRF services for payment processing systems handling 50k+ transactions per month.
- Developed payment gateway plugins in PHP for WooCommerce, PrestaShop, and Magento.
- Enforced PEP8 standards and implemented automated unit testing in CI workflows.

Intelmas - Python Developer & Linux Administrator

Jul 2018 - Oct 2019

- Delivered customer solutions using Python (Flask RESTful), MongoDB/MySQL, and PHP.
- Administered Linux servers, including scripting, debugging, directory management, and automation.
- Automated operational tasks to reduce manual workload by approximately 40%.

Education

Instituto Tecnológico de Nuevo Laredo - Master's Degree in Artificial Intelligence (in progress)

Instituto Tecnológico de Pachuca - B.Eng., Computer Systems Engineering (2018)

CBTIS #8 - IT Technician (2013)

Certifications

PCAP: Certified Associate in Python Programming (Python Institute) - Programming Essentials in Python (Cisco) - MongoDB Basics (MongoDB Inc.) - LinkedIn Learning: Advanced Python, Python Essentials, Improve Your Python Development

Skills

Languages: Python (2 & 3), JavaScript/TypeScript, Java, C/C++, PHP

Backend & APIs: FastAPI, Django, Flask, REST APIs, event-driven systems

Databases: PostgreSQL, MySQL, SQL Server, MongoDB

Cloud & DevOps: AWS (Lambda, RDS, CloudWatch, Secrets Manager), Docker, Redis, Celery, Git/GitHub

AI / LLM Tooling: LangGraph, LangSmith, OpenAI, Claude, orchestration, evaluations, guardrails

Frontend: React, Vue.js, Angular

Methods: Scrum, Kanban, Waterfall

Soft Skills: Problem-solving, clear communication, cross-team collaboration, mentorship, adaptability